

DUAL MONTE CARLO UNCERTAINTY FRAMEWORK FOR NEURAL NETWORKS: APPLICATION TO A QUANTUM VOLTAGE STANDARD

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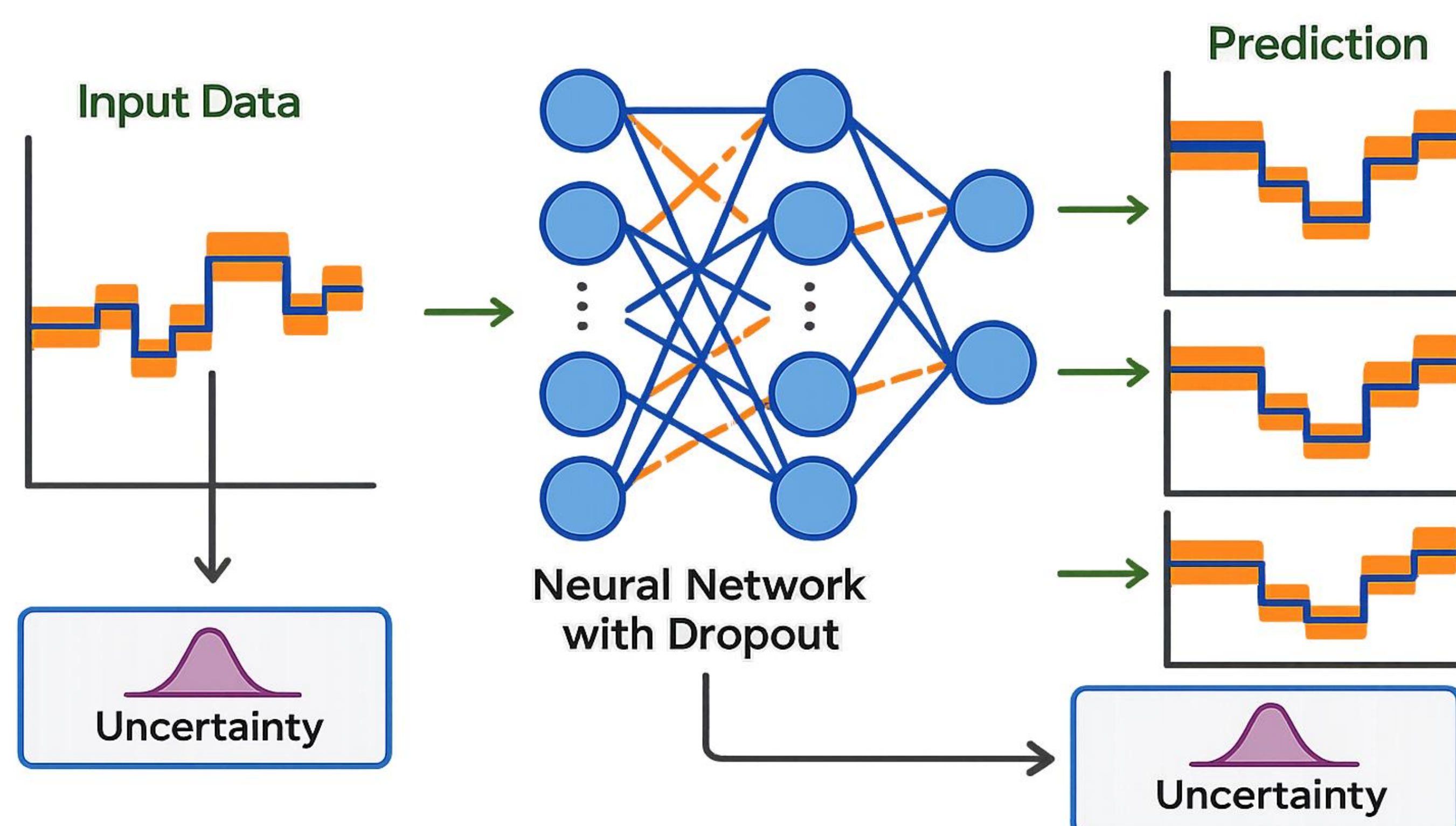
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1. Introduction

Quantum voltage standards such as the Programmable Josephson Voltage Standard (PJVS) [1] are the basis for traceable DC voltage realization with extremely low uncertainty. Data-driven models and digital twins complement physics-based descriptions of such systems for prediction, optimization, and diagnostics. Neural networks approximate nonlinear relationships and have predictive power, but introduce additional sources of uncertainty that are not explicitly addressed by classical uncertainty budgets. For metrological applications, these contributions must be quantified and reported in compliance with the GUM [2]. This work proposes the DMCUF, which combines GUM-compliant Monte Carlo [3] of measurement uncertainties with Monte Carlo Dropout [4] as an approximation of neural network uncertainty. The method is demonstrated using DC voltage measurements from a PJVS, part of a neural-network-based digital twin, developed in the scope of the EURAMET ViDiT project [5] to optimize the system, reducing setup and measurement times.

2. Method

Let $\hat{y} = f_{\omega}(x; m) + b$ denote the predictions of a trained neural network, where x is the vector of inputs, ω denotes a fixed snapshot of trained weights, m is a stochastic dropout mask, and $b = \hat{y}_{\text{ref}} - y_{\text{ref}}$ the bias of model's predictions obtained as the mean of the residuals between reference measurements y_{ref} and the predictions $\hat{y}_{\text{ref}}$. Uncertainty propagation is performed using a nested Monte Carlo scheme. In the outer loop, N input samples $x_i \sim p(x)$ are drawn from the joint input PDF. For each x_i , an inner loop performs K stochastic forward passes through the network using Monte Carlo Dropout.



The total variance of Y is decomposed using the law of total variance, $\text{Var}(Y) = E_X[\text{Var}(Y | X)] + \text{Var}_X(E[Y | X])$, and the uncertainty components are

$$u_{\text{input}} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\bar{y}_i - \bar{\mu})^2}, \quad u_{\text{model}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{1}{K-1} \sum_{j=1}^K (y_{i,j} - \bar{y}_i)^2 \right)}$$

With combined uncertainty: $u_c = \sqrt{u_{\text{input}}^2 + u_{\text{model}}^2 + u_b^2}$.

3. Results

The framework was evaluated with a 1 V DC voltage step in the 10 V PJVS systems of TÜBİTAK UME and INTI. Three neural network architectures were tested as surrogate models of the PJVS digital twin to predict DC voltage steps: a multilayer perceptron (MLP), a long short-term memory network (LSTM), and a convolutional neural network (CNN). All models were trained on the same temporally-split dataset (70% training, 20% validation, 10% test) in a one-step-ahead sequence-to-sequence setting. The DMCUF outer loop used $N = 1000$ input samples (GUM Monte Carlo) with an uncertainty estimated from the measurement series ($\sigma_{\text{input}} = 79$ nV), while the inner loop used $K = 200$ Monte Carlo Dropout forward passes (dropout rate = 40%). Reported uncertainties are standard uncertainties (coverage factor $k = 1$), and 95% empirical coverage is approximated by the interval mean $\pm 2u_c$.

Comparison of predictive performance (test set)

Model	RMSE (nV)	Bias (nV)	Coverage 95%
MLP	64	-2	96.47%
LSTM	66	-13	90.93%
CNN	69	-17	73.05%

Comparison of uncertainty components (test set)

Model	u_b (nV)	$\langle u_{\text{input}} \rangle$ (nV)	$\langle u_{\text{model}} \rangle$ (nV)	$\langle u_c \rangle$ (nV)
MLP	3	22	67	71
LSTM	3	16	50	53
CNN	3	9	33	35

4. Conclusions

A GUM-compliant dual Monte Carlo uncertainty framework has been presented for neural-network-based models and demonstrated on a PJVS application. By combining Monte Carlo input propagation with Monte Carlo Dropout, the method provides a transparent and reproducible way to quantify uncertainty contributions from both inputs and model structure. The approach is readily applicable to other metrological systems employing data-driven models.

5. References

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